Artificial Intelligence and Generative AI Applications in Neela LMS

Presenter Name: Venkata Sai Satya Mrudula Mahadasyam
Regd. No :2023000597

Dept. of AI&DS, GSCSE,GITAM, Visakhapatnam

GITAM
DEEMED TO BE UNIVERSITY

Contents

1. About the Company
2. Internship Certificate
3. Abstract
4. Objectives and Scope
5. Project Description
6. Roles and Responsibilities
7. Technology and Tools Used
8. Methodology and Implementation
9. Test Cases
10. Results & Deliverables
11. Skills and Learning Outcomes
12. Conclusion & Future Scope
13. References

GITAM
DEEMED TO BE UNIVERSITY

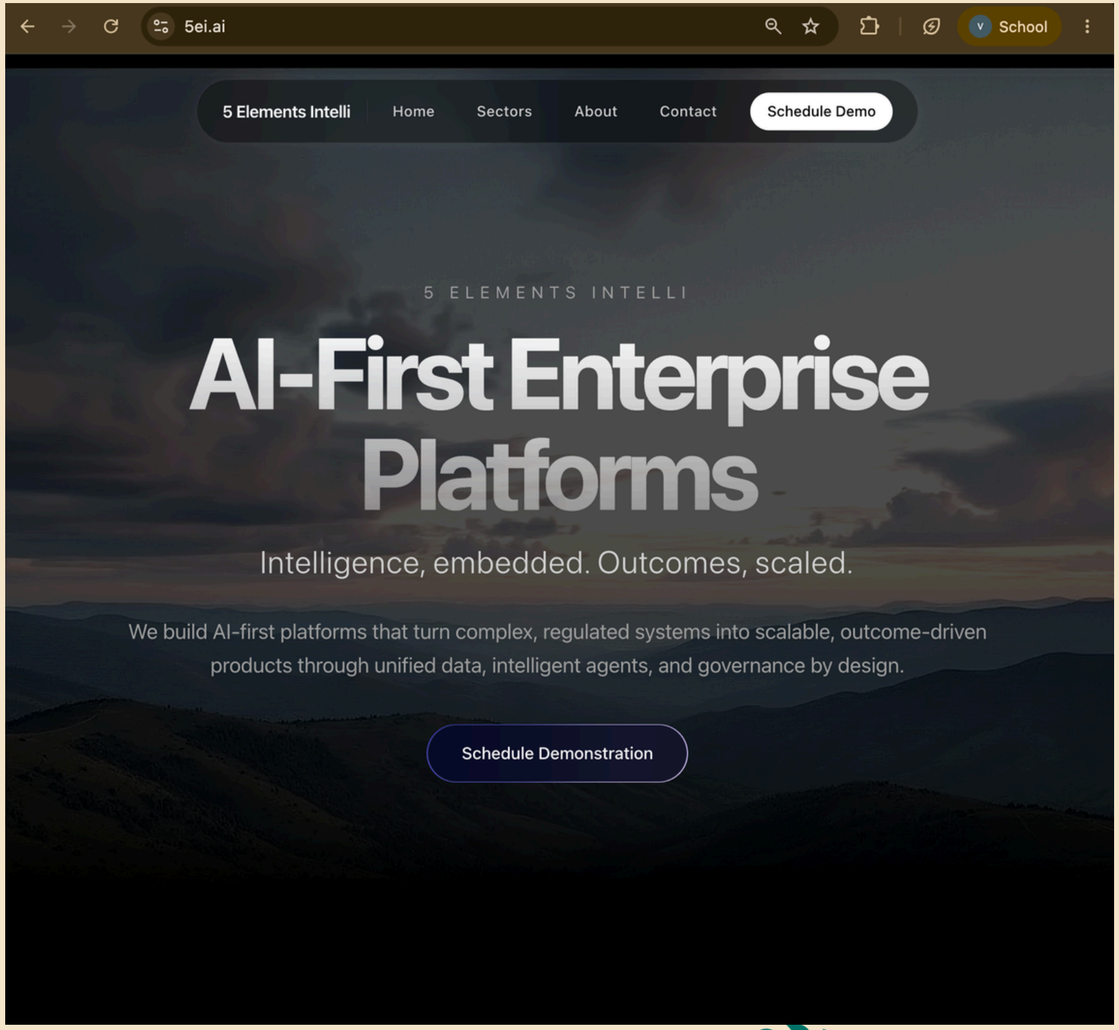
ABOUT THE COMPANY

5 ELEMENTS INTELLI PRIVATE LIMITED

- AI-first technology company building intelligent enterprise platforms.
- Uses unified data, intelligent agents, and governance by design.
- Works across Agriculture, Healthcare, Finance, and Education.
- NEELA is its education platform focused on personalized and intelligent learning.

Internship Role: AI Intern

Mentor: Shiv Buddaraju



GITAM
DEEMED TO BE UNIVERSITY

INTERNSHIP CERTIFICATE



5ei 5 ELEMENTS INTELLI PRIVATE LIMITED

20 June 2026

INTERNSHIP COMPLETION CERTIFICATE

Subject: Regarding Completion of Internship

This is to certify that Ms. Satya Mrudula Mahadasyam, a student of B.Tech CSE (AI/ML), has successfully completed an internship as an AI Intern at 5 Elements Intelli Private Limited from 5th May 2026 to 20th June 2026.

During the internship period, the intern was involved in AI-related projects and activities, including research, data analysis, model evaluation, and project documentation. The intern demonstrated dedication, professionalism, and a strong willingness to learn throughout the internship tenure.

We found the performance and conduct of the intern to be satisfactory and appreciate their contribution to the organisation.

We wish her every success in future academic and professional endeavours.

B. Sivakumar
Managing Director

This document is issued electronically on the official letterhead of the organization and does not require a physical signature for validity or authenticity.

[+91] 986 676 8368
contact@5ei.ai

1-12-3 H NO 38, SBI COLONY, PEDA WALTAIR
Visakhapatnam - 530017, Andhra Pradesh, India

GITAM
DEEMED TO BE UNIVERSITY

Geo-Tagged Photos

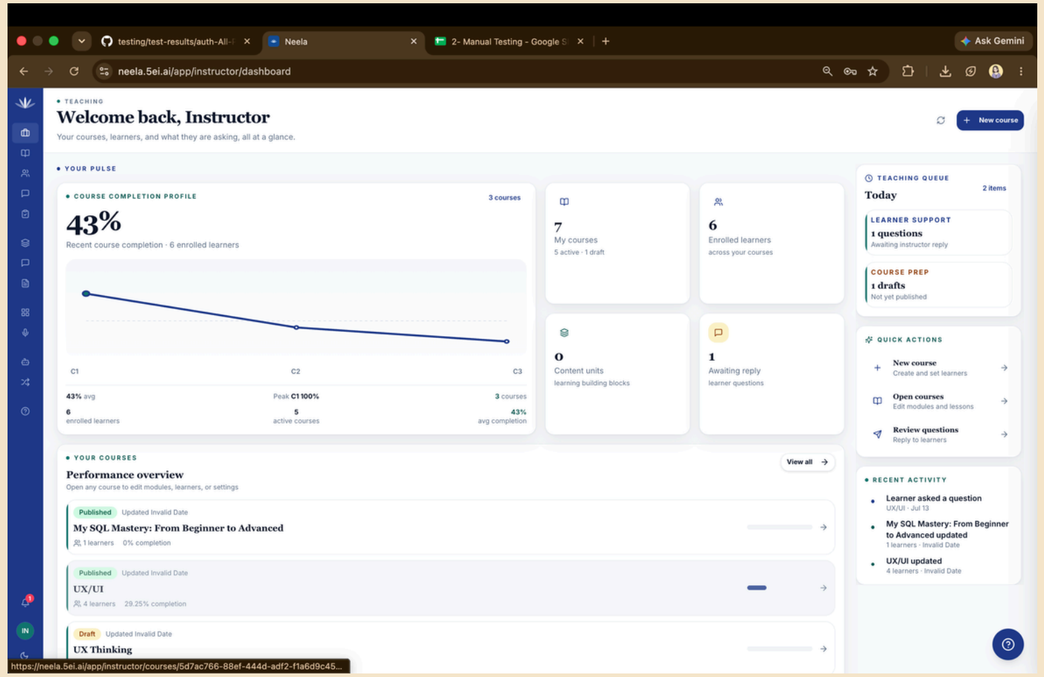




GITAM
DEEMED TO BE UNIVERSITY

ABSTRACT

- Worked as an AI Intern at 5 Elements Intelli Private Limited.
- Gained practical exposure to AI and Generative AI through the Neela LMS platform.
- Learned LLMs, RAG, embeddings, prompt engineering, and LangChain basics.
- Contributed to manual testing, test case design, bug reporting, documentation, and basic Playwright automation.
- Worked with different user personas and AI-based scenarios.



Dept. of AI&DS, GSCSE,GITAM, Visakhapatnam

GITAM
DEEMED TO BE UNIVERSITY

OBJECTIVES AND SCOPE

Objectives

- Understand AI and Generative AI in real-world applications.
- Learn LLMs, RAG, embeddings, prompt engineering, and LangChain.
- Gain practical experience in AI application testing.
- Prepare and execute test cases.
- Identify, report, and document bugs.

Scope

- Testing Neela LMS features and workflows.
- Testing different user personas.
- Evaluating AI-generated responses.
- Basic Playwright automation.
- Documentation and team collaboration.

PROJECT DESCRIPTION

NEELA LMS

Neela LMS is an AI-powered Learning Management System designed to provide a smart and interactive learning experience.

Key Features

- AI-powered Academic Assistant
- Intelligent Q&A using RAG
- Course & Learning Resource Management
- Personalized Learning Support
- Content Retrieval using Embeddings
- Secure Authentication & Role-Based Access
- Interactive User Interface

MY INTERNSHIP CONTRIBUTIONS

- Explored Neela LMS features, workflows, and different user personas.
- Performed manual testing of application features.
- Prepared and executed test cases.
- Tested AI-powered academic queries and generated responses.
- Used different prompts and inputs to evaluate AI behaviour.
- Identified and documented bugs and unexpected behaviour.
- Learned LangChain basics to understand project files and AI-related workflows.
- Gained basic exposure to Playwright automation.
- Maintained testing documentation and communicated observations with the team.

ROLES AND RESPONSIBILITIES

- AI Intern – AI feature understanding and testing.
- Manual testing of Neela LMS.
- Test case preparation and execution.
- AI response evaluation.
- Bug identification and reporting.
- Testing documentation.
- Basic Playwright automation.
- Understanding AI concepts including LLMs, RAG, embeddings, and prompt engineering.
- Collaboration with mentors and team members.

TECHNOLOGY AND TOOLS USED

AI / GenAI

- Large Language Models (LLMs)
- Generative AI
- Retrieval-Augmented Generation (RAG)
- Embeddings
- Prompt Engineering

Frameworks / Tools

- LangChain – introductory learning to understand project files and AI workflows
- Playwright – basic automation testing
- Docker & DevOps – basic exposure

Testing

- Manual Testing
- Test Case Design
- Bug Reporting
- Documentation

METHODOLOGY AND IMPLEMENTATION

Understand Feature
↓
Prepare Test Scenarios
↓
Execute Tests
↓
Observe Results
↓
Identify Issues
↓
Report & Document
↓
Retest

- Explored project files to understand the application structure.
- Learned introductory LangChain concepts to understand AI-related files.
- Tested different personas and workflows.
- Tested different queries and prompts.
- Evaluated AI-generated responses.
- Documented issues and testing observations.

TEST CASES

Feature	***Testing Focus***
AI Academic Assistant	Academic queries & responses
Intelligent Q&A / RAG	Relevant answers from learning content
Content Retrieval	Retrieval of relevant resources
Personalized Learning	User-specific learning support
Course & Resources	Access and management
Role-Based Access	Persona-specific permissions
Authentication	User access and login behaviour
Interactive UI	Navigation and user actions

RESULTS & DELIVERABLES

Results

- Contributed to QA of the Neela LMS platform.
- Tested different features, personas, and AI scenarios.
- Identified and documented issues during testing.
- Gained practical experience in AI testing and Generative AI.

Deliverables

- Test cases
- Test observations
- Bug reports
- Testing documentation
- Basic Playwright automation exposure

SKILLS AND LEARNING OUTCOMES

Technical Skills

- Generative AI & LLMs
- RAG & Embeddings
- Prompt Engineering
- LangChain Basics
- Manual Testing
- Test Case Design
- Bug Reporting
- Playwright Basics

Other Skills

- Problem Solving
- Analytical Thinking
- Communication
- Teamwork
- Documentation
- Time Management

CONCLUSION & FUTURE SCOPE

Conclusion

- Gained practical exposure to AI, Generative AI, and AI application testing.
- Connected academic concepts with a real-world AI-powered LMS.
- Improved testing, documentation, problem-solving, and communication skills.

Future Scope

- Improved AI response evaluation
- Increased test automation
- Enhanced personalization
- Better content retrieval
- Additional intelligent learning features

REFERENCES

- 5 Elements Intelli Private Limited – Internal Training & Project Resources
- Neela LMS – Project Documentation
- GitHub Documentation
- Technical resources provided by the internship mentor

Thank You

GITAM
DEEMED TO BE UNIVERSITY